

Sooraj Mohan

Embedded Systems & IoT Engineer

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PROFILE

Passionate Electronics and Communication Engineering student specializing in IoT, embedded systems, and automation. Practical experience in real-time project development using microcontrollers and sensors. Strong problem-solving, documentation, and leadership skills.

EDUCATION

B.Tech in Electronics and Communication Engineering

08/2023 – Present

Royal College of Engineering & Technology, Thrissur, Kerala, India

APJ Abdul Kalam Technological University, Kerala, India

SKILLS

Microcontrollers & Hardware: Arduino, ESP32, Embedded C, PCB Basics

Programming Languages: Verilog, C, Python

IoT & Systems: IoT System Design, Real-Time Monitoring, Sensor Integration, Automatic Control Systems, IIoT, MQTT, Modbus RTU

Web Development: React, JavaScript (Frontend), Node.js (Backend)

Soft Skills: Problem Solving, Team Collaboration, Technical Documentation, Presentation and Communication

PROFESSIONAL EXPERIENCE

Junior Hardware Engineer Intern – Volkommen Industries, Thrissur, Kerala

08/06/2026 – Present

- Assisted in the design, assembly, and testing of IoT hardware devices including Industrial IoT gateways, relay controllers, and sensor modules used in real-time industrial monitoring systems.
- Worked with Modbus RTU and MQTT communication protocols to support integration of industrial devices with IoT platforms.
- Supported hardware validation and debugging for IIoT products such as temperature sensors, data loggers, and wireless controllers deployed in industrial automation environments.
- Collaborated with the engineering team on product development and quality testing for smart monitoring solutions across energy management, cold storage, and machine health applications.

Telecommunications Intern – Bharath Sanchar Nigam Limited (BSNL)

06/2025 – 07/2025

- Observed real-time network monitoring and fault management in the Network Operations Center (NOC).
- Learned enterprise connectivity solutions, including leased lines and corporate network services.

PROJECTS

Self-Cooling Blood Bag with Real-Time Visual Expiry Indicator

04/2025 – Present

- Designed a temperature-regulated blood storage system with visual expiry indication.
- Improved blood safety and usability in emergency healthcare scenarios.

IoT Embedded Plasma Purifier

11/2025 – Present

- Built an IoT-enabled air purification system with real-time air quality monitoring.
- Implemented automatic control for efficient and intelligent operation.

Smart Helmet for Accident Detection

05/2024 – 08/2024

- Developed an IoT-based accident detection system using sensors and microcontroller.
- Enabled real-time alert generation to improve emergency response time.

ORGANISATIONS

RUDHIRASENA – APJAKTU NSS Blood Cell

06/2025 – Present

Regional Coordinator, Kerala, India

LANGUAGES

Malayalam (Native) | English (Professional) | Hindi (Intermediate) | Tamil (Basic)